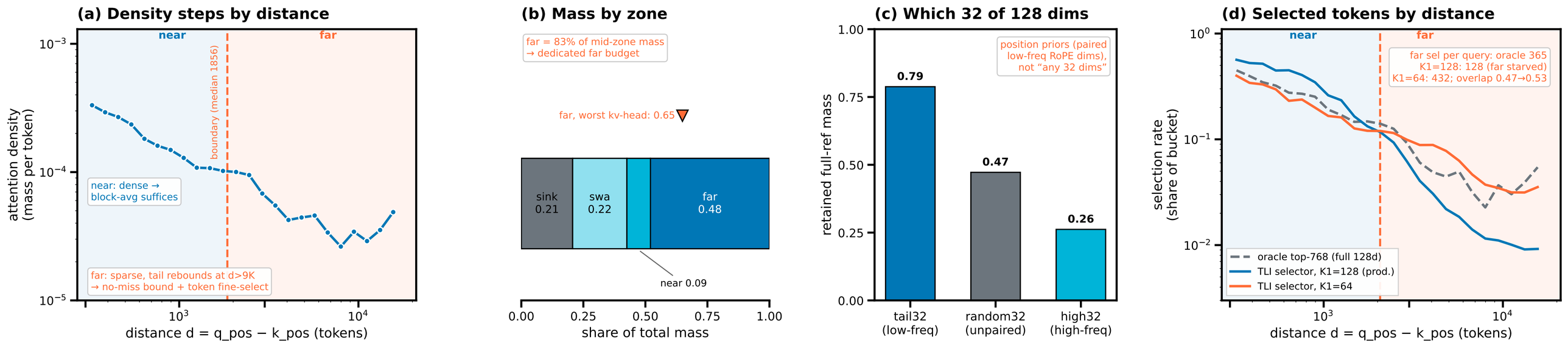
(a) Density steps by distance
near
far
boundary (median 1856)
attention density (mass per token)
near: dense →
block-avg suffices
far: sparse, tail rebounds at d>9K
→ no-miss bound + token fine-select
distance d = q_pos − k_pos (tokens)
(b) Mass by zone
far = 83% of mid-zone mass
→ dedicated far budget
far, worst kv-head: 0.65
sink
0.21
swa
0.22
far
0.48
near 0.09
share of total mass
(c) Which 32 of 128 dims
position priors (paired
low-freq RoPE dims),
not “any 32 dims”
retained full-ref mass
0.79
0.47
0.26
tail32
(low-freq)
random32
(unpaired)
high32
(high-freq)
(d) Selected tokens by distance
near
far
far sel per query: oracle 365
K1=128: 128 (far starved)
K1=64: 432; overlap 0.47→0.53
selection rate
(share of bucket)
oracle top-768 (full 128d)
TLI selector, K1=128 (prod.)
TLI selector, K1=64
distance d = q_pos − k_pos (tokens)